Philosophical logic W1

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Q1 (a) i. $V_g(\sim\phi \to \psi) = 1$ iff $V_g(\phi) = 1$ or $V_g(\psi) = 1$

($\Rightarrow$) if $V_g(\sim\phi \to \psi) = 1$, then

$$V_g(\sim\phi) = 0 \text{ or } V_g(\psi) = 1$$

which means

$$V_g(\phi) = 1 \text{ or } V_g(\psi) = 1$$

($\Leftarrow$) if $V_g(\phi) = 1$ or $V_g(\psi) = 1$

then $V_g(\sim\phi) = 0$ or $V_g(\psi) = 1$

suppose $V_g(\psi) = 1$ only

Then $V_g(\sim\phi \to \psi) = 1$

suppose $V_g(\sim\phi) = 0$ only,

then $V_g(\sim\phi \to \psi) = 1$ $\square$

ii. $V_g(\sim(\phi \to \sim\psi)) = 1$ iff $V_g(\phi) = 1$ and $V_g(\psi) = 1$

($\Rightarrow$) if $V_g(\sim(\phi \to \sim\psi)) = 1$, then

$$V_g(\phi \to \sim\psi) = 0$$

which implies $V_g(\phi) = 1$ and $V_g(\sim\psi) = 0$

$\Rightarrow V_g(\phi)=1$ and $V_g(\psi)=1$

$(\Leftarrow)$ if $V_g(\phi)=1$ and $V_g(\psi)=1$

we know that

$V_g(\sim\psi)=0$

$\therefore V_g(\phi\to\sim\psi)=0$ ↙ It's important here that $V(\phi)=1$, if that was not the case then

$\therefore V_g(\sim(\phi\to\sim\psi))=1$ $\square$ this would not follow

iii. $V_g(\sim((\phi\to\psi)\to\sim(\psi\to\phi)))=1$ iff $V_g(\phi)=V_g(\psi)$

$(\Rightarrow)$ if $V_g(\sim((\phi\to\psi)\to\sim(\psi\to\phi)))=1$,

then $V_g((\phi\to\psi)\to\sim(\psi\to\phi))=0$

$\therefore V_g(\phi\to\psi)=1$ and $V_g(\sim(\psi\to\phi))=0$

There are two cases to satisfy $V_g(\phi\to\psi)=1$

(1) $V_g(\phi)=V_g(\psi)$

and

(2) $V_g(\phi)=0$, $V_g(\psi)=1$

Consider (1): $V_g(\psi\to\phi)=1 \Rightarrow V_g(\sim(\psi\to\phi))=0$

Consider (2): $V_g(\psi \to \phi) = 0 \Rightarrow V_g(\sim(\psi \to \phi)) = 1$ ∴ ✗

So, $V_g(\phi) = V_g(\psi)$ A bit more explanation would be helpful

($\Leftarrow$) If $V_g(\phi) = V_g(\psi)$

then, $V_g(\phi \to \psi) = 1$

$$V_g(\psi \to \phi) = 1 \Rightarrow V_g(\sim(\psi \to \phi)) = 0$$

$$\Rightarrow V_g((\phi \to \psi) \to \sim(\psi \to \phi)) = 0$$ ✓

$$\therefore V_g(\sim((\phi \to \psi) \to \sim(\psi \to \phi))) = 1 \quad \square$$

(b) i. $V(\sim\phi) = 1$ iff $V(\phi) = 0$

holds because it is a primitive connective & is defined as such since '$\sim$' represents 'not'. Semantically, not ϕ holds iff ϕ does not hold ✓

ii. $V(\phi \to \psi) = 1$ iff $V(\phi) = 0$ or $V(\psi) = 1$

holds also because it is a primitive connective & is defined as such that if ϕ, then ψ is true iff ϕ is false or ψ is true ✓

iii $V(\phi \vee \psi) = 1$ iff $V(\phi) = 1$ or $V(\psi) = 1$

holds since

'$\phi \vee \psi$' is an abbreviation of '$\sim\phi \to \psi$', so $V(\phi \vee \psi) = 1$ iff $V(\sim\phi \to \psi)$

As shown in (a)(i)

$$V(\sim\phi \to \psi) = 1 \text{ iff } V(\phi) = 1 \text{ or } V(\psi) = 1$$ ✓

so $V(\phi \vee \psi) = 1$ iff $V(\phi) = 1$ or $V(\psi) = 1$

iv. $V(\phi \wedge \psi) = 1$ iff $V(\phi) = 1$ & $V(\psi) = 1$

holds since '$\phi \wedge \psi$' abbreviates '$\sim(\phi \to \sim\psi)$'

so $V(\phi \wedge \psi) = 1$ iff $V(\sim(\phi \to \sim\psi)) = 1$ ✓

since $V(\sim(\phi \to \sim\psi)) = 1$ iff $V(\phi) = 1$ and $V(\psi) = 1$

from aii),

$$V(\phi \wedge \psi) = 1 \text{ iff } V(\phi) = 1 \text{ and } V(\psi) = 1$$

v. $V(\phi \leftrightarrow \psi) = 1$ iff $V(\phi) = V(\psi)$

holds since '$\phi \leftrightarrow \psi$' abbreviates '$\sim((\phi \to \psi) \to \sim(\psi \to \phi))$'

$\therefore V(\phi \leftrightarrow \psi) = 1$ iff $V(\sim((\phi \to \psi) \to \sim(\psi \to \phi))) = 1$

since $V(\sim((\phi \to \psi) \to \sim(\psi \to \phi))) = 1$ iff $V(\phi) = V(\psi)$ from aiii)

$V(\phi \leftrightarrow \psi) = 1$ iff $V(\phi) = V(\psi)$ ✓

Halbach considers the set $\{\neg, \vee, \wedge, \rightarrow, \leftrightarrow\}$ as primitive vocabulary, while Sider only considers $\{\neg, \rightarrow\}$ as primitive. In other words, Sider defines the other three connectives in terms of his primitive vocabulary i.e. showing that $\{\neg, \rightarrow\}$ is expressively adequate

✓

Q2. (a) $\vDash_{PL} \phi \rightarrow (\psi \rightarrow \phi)$

Let A be an interpretation. For reductio, assume $V_A(\phi \rightarrow (\psi \rightarrow \phi)) = 0$ by definition of $\rightarrow$, $V_A(\phi) = 1$, $V_A(\psi \rightarrow \phi) = 0$. But then $V_A(\phi) = 0$ by definition of $\rightarrow$ and $V_A(\psi) = 1$, a contradiction $\times$ So there is no A s.t. $V_A(\phi \rightarrow (\psi \rightarrow \phi)) = 0$

✓

(b) $\vDash_{PL} (\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi))$

Let A be any PL-interpretation. For reductio, assume that

$V_A((\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi))) = 0$

Then, $V_A(((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi))) = 0$ (1) and

$V_A(\phi \rightarrow (\psi \rightarrow \chi)) = 1$ (2)

by definition of semantic clause $\rightarrow$. From (1), by definition of $\rightarrow$,

$V_A(\phi \rightarrow \chi) = 0$ (3) and $V_A(\phi \rightarrow \psi) = 1$ (4)

By definition of $\rightarrow$, $V_A(\chi) = 0$, $V_A(\phi) = 1$, $V_A(\psi) = 1$

✓

This means that $V_A(\psi \rightarrow \chi) = 0$. But (2) implies $V_A(\psi \rightarrow \chi) = 1$, a contradiction

(c) $\vDash_{PL} (\sim\psi \to \sim\phi) \to ((\sim\psi \to \phi) \to \psi)$

Let A be any PL-interpretation.

For reductio, $V_A((\sim\psi \to \sim\phi) \to ((\sim\psi \to \phi) \to \psi)) = 0$.

By the def of $\to$, $V_A(\sim\psi \to \sim\phi) = 1$ ①

$V_A((\sim\psi \to \phi) \to \psi) = 0$ ②

From ②, by def of $\to$, $V_A(\psi) = 0$, $V_A(\phi) = 1$

But then, $V_A(\sim\psi \to \sim\phi) = 0$ since $V_A(\sim\psi) = 1$, $V_A(\sim\phi) = 0$ ✓

which contradicts ①.

So the wff is true in every PL.

(d) $\phi, \phi \to \psi \vDash_{PL} \psi$

Let A be any PL-interpretation. For reductio,

let $V_A(\phi) = 1$, $V_A(\phi \to \psi) = 1$ but $V_A(\psi) = 0$

But by the definition of $\to$, ✓

$V_A(\phi) = 1$, $V_A(\phi \to \psi) = 1$ imply $V_A(\psi) = 1$

which yields a contradiction.

$\therefore$ for any PL-interpretation in which ϕ, $\phi \to \psi$ are true

ψ is also true.

Q3. Claim: if ϕ contains at most one occurrence of any sentence letter, then

$\nvDash_{PL} \phi$

We can prove an intermediate claim. Let $NC(\phi)$ denote the number of connectives in wff ϕ.

For ease, denote ϕ_k when $NC(\phi_k) = k$

Lemma: if ϕ contains at most one occurrence of any sentence letter, then there exists a PL-interpretation A s.t. $V_A(\phi) = 1$.

Base Case: Trivially, there is an A s.t. $V_A(\phi_0) = 1$. IH: if ϕ_n contains at most one sentence letter and $\exists A$ s.t. $V_A(\phi_n) = 1$, then (if ϕ_{n+1} contains at most one sentence letter, then $\exists A$ s.t. $V_A(\phi_{n+1}) = 1$)

Since $\phi_{n+1} = \phi_n \circ \psi$ and ψ is a sentence letter that does not occur in ϕ_n, by hypothesis the same A that assigns $V_A(\phi_n) = 1$

can also assign $V_A(\psi)=1$. Regardless of the binary connective $\{\wedge, \vee, \rightarrow, \Leftrightarrow\}$, $V_A(\varphi_{n+1})=1$

Proof. We prove this by induction.

Base Case: $NC(\phi_0)=0$. Then, there exists a PL-interpretation A that assigns $V_A(\phi)=0$

IH: if $NC(\varphi_n)=n$ and $\nvDash_{PL}\varphi_n$, then $NC(\phi_{n+1})=n+1$ and $\nvDash_{PL}\varphi_{n+1}$

Let us assume that $\nvDash_{PL}\varphi_n$. Then suppose φ_{n+1} has one more connective such that $\varphi_{n+1}=\phi_n\circ\psi$ where ψ is some sentence letter. Since ψ does not occur in ϕ_n, we consider the following cases:

i) $\varphi_n\wedge\psi$

By hypothesis, $\nvDash_{PL}\varphi_n$ so there is an A s.t. $V_A(\varphi_n)=0$ so $V_A(\varphi_n\wedge\psi)=0$. So, $\nvDash_{PL}\varphi_n\wedge\psi$

ii) $\varphi_n\vee\psi$

By hypothesis, $\nvDash_{PL}\varphi_n$ so there is an A s.t. $V_A(\varphi_n)=0$, and since ψ does not occur in φ_n, then A can assign $V_A(\psi)=0$

$\therefore V_A(\phi_n\vee\psi)=0 \therefore \nvDash_{PL}\Phi_n\vee\psi$

iii) $\phi_n\rightarrow\psi$.

By the lemma we proved, since by assumption φ_n has at most one occurrence of any sentence letter, there is an A s.t. $V_A(\varphi_n)=1$. Since ψ does not occur in ϕ_n, A can assign $V_A(\psi)=0$. $\therefore$ $V_A(\varphi_n\rightarrow\psi)=0$, $\therefore \nvDash_{PL}\varphi_n\rightarrow\varphi$

iv) $\phi_n\Leftrightarrow\psi$.

Again, by the lemma, there is an A s.t. $V_A(\varphi_n)=1$. Since ψ does not occur in φ_n, A can assign $V_A(\psi)=0$.

$\therefore V_A(\phi_n\Leftrightarrow\psi)=0$, $\therefore \nvDash_{PL}\phi_n\Leftrightarrow\psi$

To do proof by induction you also need to consider the negation case. In fact, since $\{\neg, \rightarrow\}$ are adequate, you only need to consider these two cases. This is an easy fix tho, and the rest of the proof is good!

✓

So, $\nvDash_{PL}\varphi_{n+1}$. IH is proven

So, if ϕ has at most one occurrence of any sentence letter, then $\nvDash_{PL}\phi$

✓

Q4. (a) $V_A(\phi\downarrow\psi)=1$ iff $V_A(\phi)=0$ and $V_A(\psi)=0$

↓	1	0
1	0	0
0	0	1

'A nor B' is only true if 'A' and 'B' are both false. 'nor' is the opposite of 'or'

(b) $\phi \downarrow \phi$. If $V_A(\phi) = 0$,

abbreviates $V_A(\phi \downarrow \phi) = 1$

$\sim\phi$ If $V_A(\phi) = 1$,

$V_A(\phi \downarrow \phi) = 0$

which are the same as $\sim\phi$

$\phi \to \psi$ is the same as $\neg\phi \vee \psi$

$\Rightarrow (\phi \downarrow \phi) \vee \psi$

$\phi \vee \psi$ is just $\neg(\phi \downarrow \psi)$ which is

just $(\phi \downarrow \psi) \downarrow (\phi \downarrow \psi)$

so $((\phi \downarrow \phi) \downarrow \psi) \downarrow ((\phi \downarrow \phi) \downarrow \psi)$ expresses $\phi \to \psi$

(c) $\{\downarrow\}$ is adequate since it can symbolize the truth functions symbolized by $\{\neg, \to\}$ as shown in (b)

since $\phi \wedge \psi$ abbreviates $\sim(\phi \to \sim\psi)$

$\phi \vee \psi$ abbreviates $\sim\phi \to \psi$

$\phi \leftrightarrow \psi$ abbreviates $\sim((\phi \to \psi) \to \sim(\psi \to \phi))$

$\downarrow$ can express $\{\wedge, \vee, \leftrightarrow\}$ as well through expressing the same truth functions of $\{\neg, \to\}$

Q5. (a) i. $P, (P \to Q) \vDash_K Q$ is true.

Let A be a K-interpretation. By assumption $V_A(P) = 1$, $V_A(P \to Q) = 1$. For reductio, assume $V_A(Q) = \#$ or $V_A(\phi) = 0$. But if $V_A(P) = 1$, $V_A(P \to Q) = 1$, then $V_A(Q) = 1$ by def. of $\to$, a contradiction ii $\models_K P \to ((P \to Q) \to Q)$

P	Q	P	$\to$	$(($ P	$\to$	$Q)$	$\to$	$Q)$
		1	0	1	?	0	0	0
		1	#	1	#	#	#	#

Not true. When $V_A(P)=1$, $V_A(Q)=\#$,

then $V_A(P \to ((P \to Q) \to Q)) = \#$

so $\not\models_K P \to ((P \to Q) \to Q)$ as $\exists A$ s.t.

$V_A(P \to ((P \to Q) \to Q)) \neq 1$

iii $P, \sim P \models_K Q$ holds true

since $\Gamma \models_K \phi$ iff there is no A s.t. $V_A(\phi) = 0$ and $V_A(\psi) = 1, \forall \psi \in \Gamma$

since if $V_A(P) = 1$, $V_A(\sim P) = 0$ & vice versa

there is no A s.t. $V_A(\psi) = 1, \forall \psi \in \Gamma$

iv. $\models_K (P \wedge \sim P) \to Q$

P	Q	$(P$	$\wedge$	$\sim P)$	$\to$	Q
		#	#	#	#	#

so $\not\models_K (P \wedge \sim P) \to Q$

(b) i. $P, P \to Q \models_L Q$ holds since

let $V_A(P) = 1$, $V_A(P \to Q) = 1$.

Assume that $V_A(Q) \neq 1$.

But by the def of $\to$, if $V_A(Q) = \#$

then $V_A(P \to Q) = \#$

and if $V_A(Q) = 0$,

then $V_A(P \to Q) = 0$ $\times$

✓

so $P, P \to Q \models_L Q$

ii $\models_L P \to ((P \to Q) \to Q)$

P Q	$P \to ((P \to Q) \to Q)$
	1 0 1 ? 0 0 0
	# # # # 0 0 0

So there is an A assigning $V_A(Q) = 0$, $V_A(P) = \#$

s.t. $V_A(\underset{\#}{P} \to ((\underset{\#}{P} \underset{\#}{\to} \underset{Q}{Q}) \underset{\#}{\to} \underset{0}{Q}) = \#$ 1

$\therefore \not\models_L P \to ((P \to Q) \to Q)$

No, this evaluation gives you $V_A(P \to ((P \to Q) \to Q)$ $= 1$

$V(P \to Q) = \# \Rightarrow V((P \to Q) \to Q) = \#$ and so

$V(P \to ((P \to Q) \to Q) = 1$ as $V(\# \to \#) = 1$

iii $P, \sim P \models_{\L} Q$ holds

since $\Gamma \models_{\L} \phi$ iff $\neg\exists A$ s.t. $V_A(\phi) \neq 1$

and $V_A(\psi) = 1, \forall\psi \in \Gamma$

since if $V_A(P) = 1$ then $V_A(\sim P) = 0$ & vice versa,

there is no A s.t. $V_A(\psi) = 1, \forall \psi \in \Gamma$

iv $\models_{\L} (P \wedge \sim P) \rightarrow Q$

P	Q	$(P$	$\wedge$	$\sim$	$P)$	$\rightarrow$	Q
		1	1	1	?	0	0
		1	?	0	1	#	#
		#	#	#	#	#	0

so $\not\models_{\L} (P \wedge \sim P) \rightarrow Q$

since $\exists A$ s.t. $V_A((P \wedge \sim P \rightarrow Q) = \#$

(c) i. $P, P \rightarrow Q \models_{LP} Q$

Assume $V_A(P) \neq 0$, $V_A(P \rightarrow Q) \neq 0$

let $V_A(Q) = 0$

Suppose $V_A(P) = 1$,

then $V_A(P \rightarrow Q) = 0$ ✗

Suppose $V_A(P) = \#$,

then $V_A(P \rightarrow Q) = \#$

so, $P, P \rightarrow Q \not\models_{LP} Q$

ii. $\models_{LP} P \to ((P \to Q) \to Q)$

P	Q	$P \to ((P \to Q) \to Q)$
		1 0 1 ? 0 0 0

✓

so $\models_{LP} P \to ((P \to Q) \to Q)$

iii. $P, \sim P \models_{LP} Q$

Assume $V_A(P) \neq 0, V_A(\sim P) \neq 0$

and let $V_A(Q) = 0$

if $V_A(P) = V_A(\sim P) = \#$

while $V_A(Q) = 0$ ✓

so, $P, \sim P \not\models_{LP} Q$

iv. $\models_{LP} (P \wedge \sim P) \to Q$

P	Q	$(P \wedge \sim P) \to Q$
		1 1 ? 1 0 0

✓

so $\models_{LP} (P \wedge \sim P) \to Q$

Q6 (a) let $\# = \frac{1}{2}$, so $1 > \# > 0$

from LfP,

$$LV(\phi \to \psi) = \begin{cases} 1 & \text{if } LV(\phi) = 0 \text{ or } LV(\psi) = 1 \\ & \text{or } LV(\phi) = LV(\psi) = \# \\ 0, & \text{if } LV(\phi) = 1 \text{ and } LV(\psi) = 0 \\ \# & \text{otherwise} \end{cases}$$

Assume $LV(\phi) = 0$ or $LV(\psi) = 1$ or $LV(\phi) = LV(\psi) = \#$

Suppose $LV(\phi) = 0$, then $LV(\psi) = 0$ or $\#$ or 1

so $LV(\psi) \geq LV(\phi)$

Suppose $LV(\psi) = 1$, then $LV(\phi) = 0$ or $\#$ or 1

so $LV(\psi) \geq LV(\phi)$

Suppose $LV(\psi) = LV(\phi) = \#$

then $LV(\psi) = LV(\phi) \Rightarrow LV(\psi) \geqslant LV(\phi)$

so $LV(\psi) \geqslant LV(\phi)$

Assume $LV(\psi) \geqslant LV(\phi)$,

then if $LV(\psi) = 1$, $LV(\phi) = 1$ or $\#$ or 0

$LV(\psi) = \#$, $LV(\phi) = \#$ or 0

$LV(\psi) = 0$, $LV(\phi) = 0$

: if $LV(\phi) = 0$, $LV(\psi) = 1$ or $\#$ or 0 ✓

$LV(\phi) = \#$, $LV(\psi) = 1$ or $\#$

$LV(\phi) = 1$, $LV(\psi) = 1$

which implies that $LV(\phi) = 0$ or $LV(\psi) = 1$ or

$LV(\phi) = LV(\psi) = \#$

Hence if $LV(\psi) \geqslant LV(\phi)$,

then $LV(\phi \to \psi) = 1$

Assume that $LV(\psi) < LV(\phi)$

then we know that $LV(\phi) \neq 0$, $LV(\psi) \neq 1$

let $LV(\phi) = 1$, $LV(\psi) = 0$

which implies $LV(\phi \to \psi) = 0$

which is the same as $LV(\phi \to \psi) = 1 - (LV(\phi) - LV(\psi))$

let $LV(\phi) = 1$, $LV(\psi) = \#$

which implies $LV(\phi \to \psi) = \#$

which is $LV(\phi \to \psi) = 1 - (LV(\phi) - LV(\psi))$

let $LV(\phi) = \#$, $LV(\psi) = 0$

which implies $LV(\phi \to \psi) = \#$, ✓

which is $LV(\phi \to \psi) = 1 - (LV(\phi) - LV(\psi))$

So, if $LV(\psi) < LV(\phi)$,

$LV(\phi \to \psi) = 1 - (LV(\phi) - LV(\psi))$

(h) let l be a presicification of A. for any α, $l(\alpha) = \begin{cases} 1 & \text{if } V_A(\alpha) = \# \\ V_A(\alpha) & \text{otherwise} \end{cases}$

$l'(\alpha) = \begin{cases} 0 & \text{if } V_A(\alpha) = \# \\ V_A(\alpha) & \text{otherwise} \end{cases}$

if $KV_A(\phi \to \psi) = 1$,

then we have two cases

1) $V_A(\phi) = 1$ or $\#$ or 0 and $V_A(\psi) = 1$

in which case $V_l(\psi) = 1$, for every l of A

so $SV_A = (\psi) = 1$,

$\therefore SV_A(\phi \to \psi) = 1$

2) $V_A(\phi) = 0$ and $V_A(\psi) = 1$ or $\#$ or 0

in which case $V_l(\phi) = 0$ for $\forall l$ of A

so $SV_A(\phi) = 0$

$\therefore SV_A(\phi \to \psi) = 1$

let $KV_A(\phi \to \psi) = \#$

then

1) $V_A(\phi) = V_A(\psi) = \#$

since ϕ & ψ are distinct

there is a l that assigns

$V_l(\phi) = 0$, $V_l(\psi) = 1$

so $V_l(\phi \to \psi) = 1$

there is a l' that assigns

$V_{l'}(\phi) = 1$, $V_{l'}(\psi) = 0$, so $V_l(\phi \to \psi) = 0$

$\therefore SV_A(\phi \to \psi) = \#$

2) $V_A(\phi)=1$ and $V_A(\psi)=\#$

there is a ℓ s.t.

$V_\ell(\psi)=0 \Rightarrow V_\ell(\phi\to\psi)=0$

and a ℓ' s.t.

$V_\ell(\psi)=1 \Rightarrow V_\ell(\phi\to\psi)=1$

$\therefore SV_A(\phi\to\psi)=\#$

3) $V_A(\phi)=\#$ and $V_A(\psi)=0$

there is a ℓ s.t.

$V_\ell(\phi)=1 \Rightarrow V_\ell(\phi\to\psi)=0$

there is a ℓ' s.t.

$V_{\ell'}(\phi)=0 \Rightarrow V_\ell(\phi\to\psi)=1$

$\therefore SV_A(\phi\to\psi)=\#$

let $KV_A(\phi\to\psi)=0$

and since $V_A(\phi)=1$, $V_A(\psi)=0$,

then $V_\ell(\phi)=1$ and $V_\ell(\psi)=0$ for all ℓ of A

$\therefore SV_A(\phi\to\psi)=0$ ✓

$\therefore KV_A(\phi\to\psi)=SV_A(\phi\to\psi)$

This still holds if ϕ & ψ are not atomic but distinct since ϕ & ψ can still be assigned truth values without impinging on the other

Does not hold if not distinct

Suppose $KV_A(\phi \to \phi) = \#$

implies $V_A(\phi) = \#$

let $V_\ell(\phi) = 1$

then $V_\ell(\phi \to \phi) = 1$

let $V_{\ell'}(\phi) = 0$

then $V_{\ell'}(\phi \to \phi) = 1$

so $SV_A(\phi \to \phi) = 1$

$\therefore KV_A(\phi \to \psi) \neq SV_A(\phi \to \psi)$

trivially when $\phi = \psi$

(c) Claim: $P \models_{PL} \phi$ iff $P \models_S \phi$

($\Rightarrow$) let us assume $P \not\models_S \phi$

then there is a A s.t. $SV_A(\phi) = 0$ and

$SV_A(\psi) = 1$ for all $\psi \in P$

if $SV_A(\phi) = 0$, then $V_\ell(\phi) = 0$ for all presicifications ℓ of A

· if $V_\ell(\phi) = 0$, then $V_A(\phi) = 0$ or $V_A(\phi) = \#$

Likewise, if $SV_A(\psi) = 1$, then $V_\ell(\psi) = 1$ for all ℓ of A

if $V_\ell(\psi) = 1$, then $V_A(\psi) = 1$ or $V_A(\psi) = \#$

what this means is that it is possible to construct a bivalent PL-interpretation α s.t. $V_\alpha(\psi) = 1, \forall \psi \in \Gamma$ and $V_\alpha(\phi) = 0$

so, $\Gamma \nvDash_{PL} \phi$

By contraposition, if $\Gamma \vDash_{PL} \phi$, then $\Gamma \vDash_S \phi$

$(\Leftarrow)$ let us assume $\Gamma \nvDash_{PL} \phi$.

Then, $\exists A$ s.t. $V_A(\phi) = 0$ and $V_A(\psi) = 1, \forall \psi \in \Gamma$.

Then, $V_\ell(\phi) = 0$ for all ℓ of $\bar{A}$

$\therefore SV_A(\phi) = 0$

and $V_\ell(\psi) = 1$ for all ℓ of $\bar{A}$, $\forall \psi \in \Gamma$

$\therefore SV_A(\psi) = 1, \forall \psi \in \Gamma$ /

$\therefore \Gamma \nvDash_S \phi$

By contraposition, if $\Gamma \vDash_S \phi$, then $\Gamma \vDash_{PL} \phi$

$\therefore \Gamma \vDash_{PL} \phi$ iff $\Gamma \vDash_S \phi$

Q7. (a) a ϑ-interpretation is intuitively faithful since it captures the point of the Sorites argument to have the first instance / Base Case P_0 satisfy the property s.t. $\vartheta(P_0) = 1$ and the final instance P_{100} not satisfy the property s.t. $\vartheta(P_{100}) = 0$

It also is intuitive because it says that the next instance has the property only if the instance before it had it, and conversely, if the nth instance did not have the property, the next instance does not have it ✓

It is faithful because it conforms to our intuitions about the argument

(b) let $1 > \# > 0$

if an ϑ-interpretation is faithful,

then by def, $\vartheta(P_0) = 1$ and

$$\vartheta(P_{100}) = 0$$

So, $\vartheta(P_0) \geq \vartheta(P_1)$

Now, if ϑ is faithful, then

if $\vartheta(P_{n+1}) = 1$, then $\vartheta(P_n) = 1$ and if $\vartheta(P_n) = 0$, then $\vartheta(P_{n+1}) = 0$

we can't pick, because this has to hold for all faithful interpretations

This means we can choose a P_k s.t. $0 < k < 100$ s.t. $\vartheta(P_k) = \#$ since it does not violate either since we can assume that $\vartheta(P_i) = 1$, for $0 \leq i < k$, as if we assume that for some i, $\vartheta(P_i) = 0$, then

$$1 = \vartheta(P_0) \geq \vartheta(P_1) \geq \dots \geq \vartheta(P_{100}) = 0$$

is trivially satisfied

from $g(P_k) = \#$, we know that $g(P_{k+1}) \neq 1$ and can only be $\#$ or 0, and applies for any P_{k+j}, $k+j \leq 100$

$\therefore$ we can choose some j where $k+j \leq 100$ without loss s.t.

$$g(P_{k+j}) = 0$$

✗

which implies that

$$1 = g(P_0) \geqslant g(P_1) \geqslant \ldots \geqslant g(P_{100}) = 0$$

(c) i. α does not hold for PL

Assume that $V_A(P_0) = V_A(P_0 \to P_1) = \ldots = V_A(P_{99} \to P_{100}) = 1$

and AFC that $V_A(P_{100}) = 0$

Then $V_A(P_{99}) = 0$

$V_A(P_{98}) = 0$ ✓

$\vdots$

$V_A(P_0) = 0$

which contradicts the assumption

β holds for PL since for any faithful interpretation,

$$1 = g(P_0) \geqslant g(P_1) \geqslant \ldots \geqslant g(P_{100}) = 0$$

and if $P_0, P_0 \to P_1, \ldots, P_{99} \to P_{100} \models_{PL} P_{100}$

then it must be that for any $\mathcal{I}$, there is a k where $0<k<100$ s.t.

$$\mathcal{I}(P_k) = 0$$ why?

so one of the premises are false

ii. α does not hold for Kleene since we proved earlier that modus ponens holds for Kleene, so P_{100} must be a semantic consequence ✓

β does not hold for Kleene since there could be some $\mathcal{I}$ interpretation that assigns ✓

$$\mathcal{I}(P_1) = 1 \text{ and } \mathcal{I}(P_2) = \dots \mathcal{I}(P_{99}) = \# \text{ and } \mathcal{I}(P_{100}) = 0$$

iii α does hold for LP since we proved that modus ponens fails for LP ✓

β holds for LP since

$$P_0, P_0 \to P_1, \dots, P_{99} \to P_{100} \models_{LP} P_{100}$$

Did we not prove that MP fails in LP?

if there was A s.t. $V_A(P_0) \neq 0, \dots, V_A(P_{99} \to P_{100}) \neq 0$ but $V_A(P_{100}) = 0$ ✗

Notice that simply assigning # to some or all P_k would provide such an A, as then $V_A(P_n \to P_{n+1}) = \#$ if $V_A(P_n) = \#$ or $V_A(P_{n+1}) = \#$ since if $V_A(P_n) = \#$, $V_A(P_{n+1}) \neq 1$ under a faithful interpretation and if $V_A(P_{n+1}) = \#$, then $V_A(P_n) \neq 0$ under a faithful interpretation

so, it must be that one of the premises at least, is false under every faithful interpretation to preserve semantic consequence

(d) Upholding α violates modus ponens which is an intuitive principle ✓

(e) Upholding β implies there is some sort of contradiction amongst the premises even though the premises seem innocent of contradiction ✓

(f) α does not hold. Let us assume that

$V_A(\Delta P_0) = 1$, $V_A(\Delta P_0 \to \Delta P_1) = 1, \ldots, V_A(\Delta P_{99} \to \Delta P_{100}) = 1$

but $V_A(\Delta P_{100}) = 0$

Then $V_A(\Delta P_{99}) = 0$ ✓

$\vdots$

$V_A(\Delta P_0) = 0$ ⨳

so $\Delta P_0, \Delta P_0 \to \Delta P_1, \ldots, \Delta P_{99} \to \Delta P_{100} \models_K \Delta P_{100}$

β holds, since if we attempt to assign $V_A(P_k) = \#$

then $V_A(\Delta P_k) = 0$

$\therefore$ if A is a faithful interpretation, $V_A(\Delta P_{100}) = 0$

and $V_A(\Delta P_0) = 1$,

Consider some arbitrary premise $\Delta P_k \to \Delta P_{k+1}$

if we assign $V_A(P_k) = \#$, $V_A(\Delta P_k) = 0$

but then all premises after $\Delta P_k \to \Delta P_{k+1}$ will be true, but ΔP_{100} false

so, we must assign $V_A(P_{k+1}) = \#$, which means $V_A(\Delta P_{k+1}) = 0$

and $V_A(P_k) = 1$, so $V_A(\Delta P_k \to \Delta P_{k+1}) = 0$, to preserve

$\Delta P_0, \Delta P_0 \to \Delta P_1, \ldots, \Delta P_{99} \to \Delta P_{100} \models_K \Delta P_{100}$

So, at least one premise must be false under any faithful interpretation

(g) It seems that if I am right about α and β in parts (d) & (e), then

Kleene's trivalent semantic best captures our intuitions about the Sorites argument, namely, that we want to allow for modus ponens to be an acceptable logical rule, but also want to deny that '100-years young is true' without saying that the premises entail a contradiction. This is achieved by Kleene's logic via 'gappy' semantics and allowing the intermediate premises to be vague under a faithful interpretation.

Classical PL clearly fails for this job because it entails that there is some contradiction to preserve semantic consequence while LP does not even have modus ponens as a logical rule.

Lastly, in (g) we see that the inclusion of a 'determinately' one-place connective disrupts what is appealing about Kleene's extension since it must uphold β. This is because the Δ operator does not allow for second-order vagueness — that it is vague if something is vague. So, we are forced to say if ΔP_k is true or false, even though something like ΔP_{20} may be vague if it is definitely the case that a 20-year old is young.